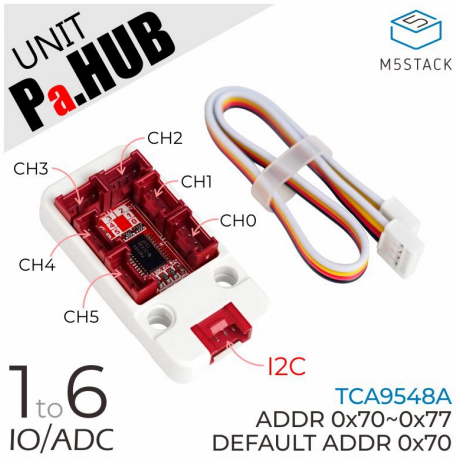
Unit PaHub

SKU:U040







Description

Unit PaHub is an I2C device splitter that can expand a single I2C HY2.0-4P interface to six channels, allowing the connection of multiple I2C devices with the same address (by controlling the polling of different channels to enable coexistence of devices with the same address). It features an embedded TCA9548A-I2C multi-channel switch IC, supporting the expansion of up to 6 groups of I2C devices.

Features

- I2C HY2.0-4P PORTA expansion
- 2 x LEGO-compatible holes
- Allows multiple nesting
- 1-6 expansion

Includes

- 1 x Unit PaHub
- 1 x HY2.0-4P Grove cable (20cm)

Applications

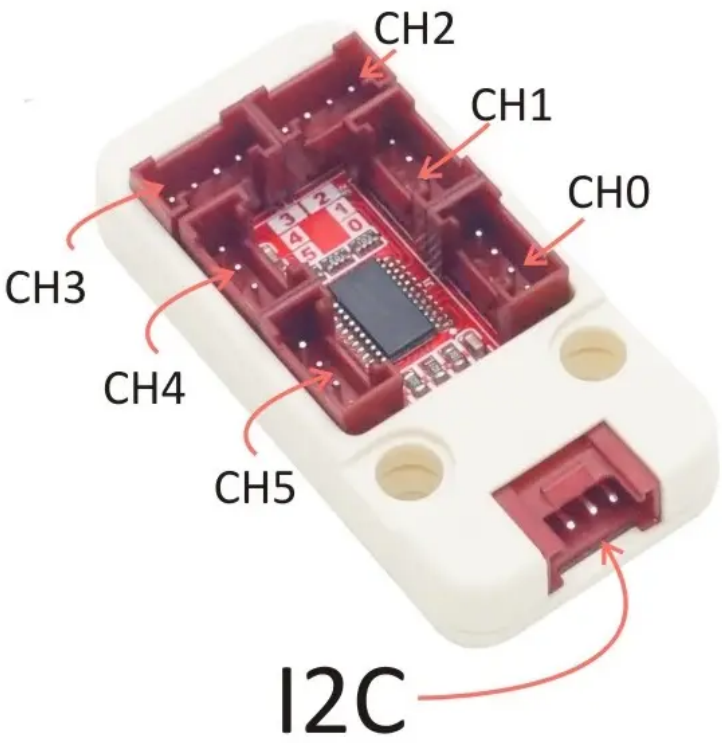
- I2C expansion

Specifications

Specification	Parameter
Protocol	I2C @0x70 (modifiable via resistors A0, A1, A2)
Product Size	48.0 x 24.0 x 10.8mm
Product Weight	6.7g
Package Size	138.0 x 93.0 x 11.8mm
Gross Weight	12.0g

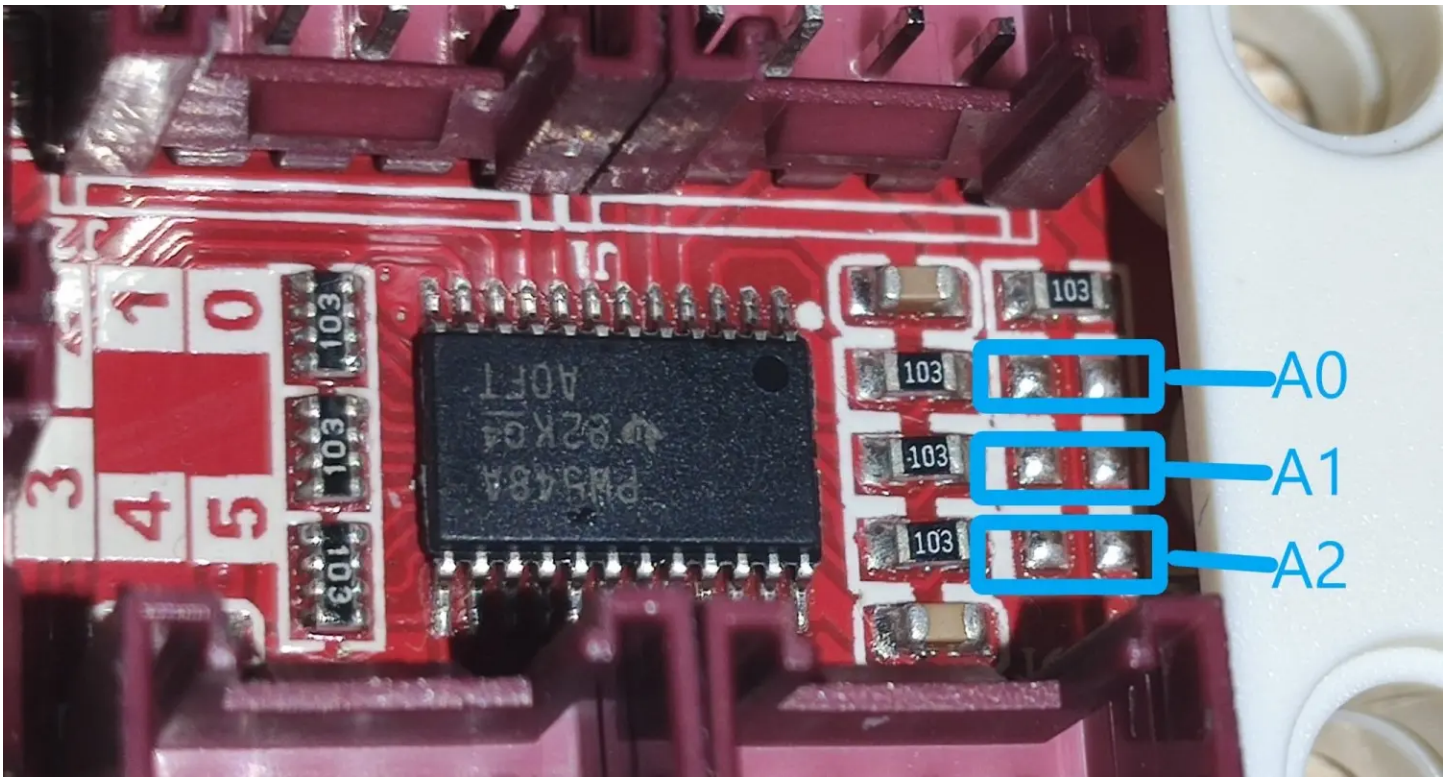
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Note: During operation, the splitter allows devices with the same address to coexist by controlling the polling of different channels. Please pay attention to the channel sequence during programming.



According to the reference schematic and the TCA9548A datasheet, this Unit can modify the I2C address of the device by controlling the level combination of the A0~A2 pins. (The default address is 0x70).

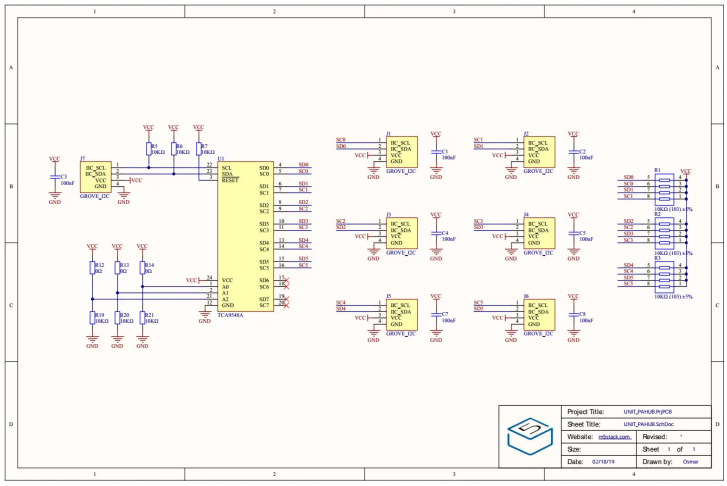
The PCB of the Unit has three reserved positions for surface-mount resistors, labeled A0-A2, as shown below.



After soldering a 0-ohm resistor, the corresponding pin will change from low level to high level. The pin level combinations and their corresponding I2C addresses are shown in the table below.

INPUTS			I^2C BUS SLAVE ADDRESS
A2	A1	A0	
L	L	L	112 (decimal), 70 (hexadecimal)
L	L	H	113 (decimal), 71 (hexadecimal)
L	H	L	114 (decimal), 72 (hexadecimal)
L	H	H	115 (decimal), 73 (hexadecimal)
H	L	L	116 (decimal), 74 (hexadecimal)
H	L	H	117 (decimal), 75 (hexadecimal)
H	H	L	118 (decimal), 76 (hexadecimal)
H	H	H	119 (decimal), 77 (hexadecimal)

Schematics



PinMap

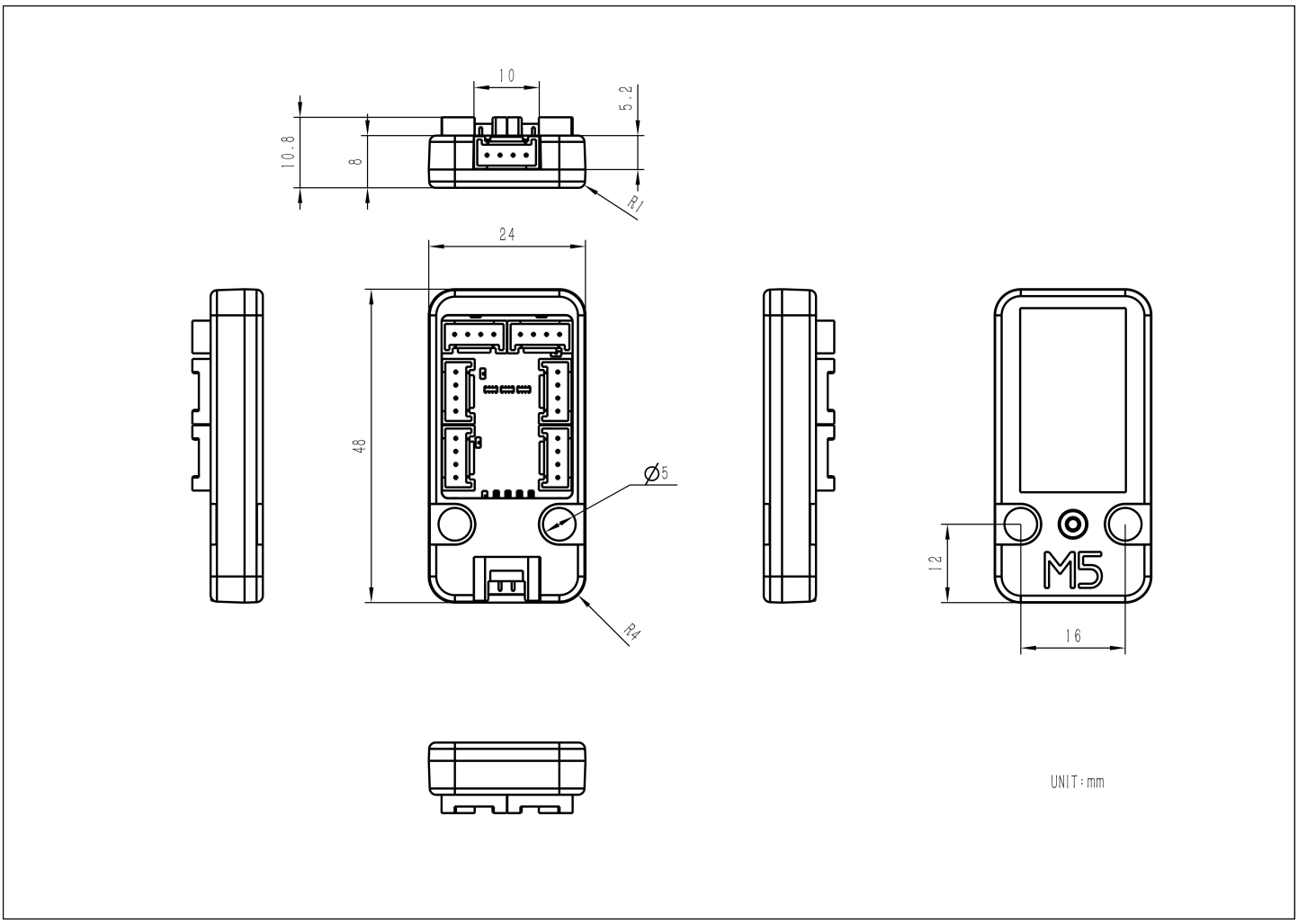
Unit PaHub

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Datasheets

- TCA9548A

Model Size



Softwares

Arduino

- Unit PaHub Test Example

UiFlow1

- UiFlow1 Docs

EasyLoader

Easyloader	Download Link	Remarks
Unit PaHub Test Easyloader	download	/